

CS 440: Introduction to Artificial Intelligence

Assignment 3 — Probabilistic Reasoning

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1 Problem 1 (15 points): Bayesian Network

The Bayesian network contains five Boolean variables A through E . A , B , and C are root nodes; D depends on A and B ; E depends on B and C .

$$P(A = T) = 0.2, \quad P(B = T) = 0.5, \quad P(C = T) = 0.8$$

A	B	$P(D = T \mid A, B)$	B	C	$P(E = T \mid B, C)$
F	F	0.9	F	F	0.2
F	T	0.6	F	T	0.4
T	F	0.5	T	F	0.8
T	T	0.1	T	T	0.3

The joint probability factorizes as:

$$P(A, B, C, D, E) = P(A) \cdot P(B) \cdot P(C) \cdot P(D \mid A, B) \cdot P(E \mid B, C)$$

(a) All five variables simultaneously true

$$\begin{aligned} P(A, B, C, D, E) &= P(A) \cdot P(B) \cdot P(C) \cdot P(D \mid A=T, B=T) \cdot P(E \mid B=T, C=T) \\ &= 0.2 \times 0.5 \times 0.8 \times 0.1 \times 0.3 = \boxed{0.0024} \end{aligned}$$

The probability that all five Boolean variables are simultaneously true is **0.0024**.

(b) All five variables simultaneously false

$P(D=F \mid A=F, B=F) = 1 - 0.9 = 0.1$ and $P(E=F \mid B=F, C=F) = 1 - 0.2 = 0.8$:

$$P(\neg A, \neg B, \neg C, \neg D, \neg E) = 0.8 \times 0.5 \times 0.2 \times 0.1 \times 0.8 = \boxed{0.0064}$$

The probability that all five Boolean variables are simultaneously false is **0.0064**.

(c) $P(\neg A \mid B, C, D, E)$

Using the normalization trick with $\alpha = \frac{1}{P(A, B, C, D, E) + P(\neg A, B, C, D, E)}$:

$$P(\neg A \mid B, C, D, E) = \frac{P(\neg A, B, C, D, E)}{P(A, B, C, D, E) + P(\neg A, B, C, D, E)}$$

The numerator uses $P(D=T \mid A=F, B=T) = 0.6$ from the CPT:

$$P(\neg A, B, C, D, E) = 0.8 \times 0.5 \times 0.8 \times 0.6 \times 0.3 = 0.0576$$

$$P(\neg A \mid B, C, D, E) = \frac{0.0576}{0.0024 + 0.0576} = \frac{0.0576}{0.0600} = \boxed{0.96}$$

Given that B , C , D , and E are all true, there is a **96% probability that A is false**. The high prior $P(\neg A) = 0.8$ combined with the favorable $P(D=T \mid \neg A, B=T) = 0.6$ drives this result.

2 Problem 2 (15 points): Burglary Network

(a) $P(\text{Burglary} \mid \text{JohnsCalls} = T, \text{MaryCalls} = T)$

$$P(B) = 0.001, \quad P(E) = 0.002$$

B	E	$P(A=T \mid B, E)$	$P(A=F \mid B, E)$		A	$P(J=T \mid A)$
T	T	0.95	0.05		T	0.90
T	F	0.94	0.06		F	0.05
F	T	0.29	0.71			
F	F	0.001	0.999			
					A	$P(M=T \mid A)$
					T	0.70
					F	0.01

With $\alpha = \frac{1}{P(J,M)}$, marginalizing over E and A :

$$P(B \mid J, M) = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J=T \mid A) P(M=T \mid A)$$

The inner sum over A evaluates to:

B	E	$A=T$ term	$A=F$ term	$f(B, E)$
T	T	$0.95 \times 0.9 \times 0.7 = 0.598500$	$0.05 \times 0.05 \times 0.01 = 0.000025$	0.598525
T	F	$0.94 \times 0.9 \times 0.7 = 0.592200$	$0.06 \times 0.05 \times 0.01 = 0.000030$	0.592230
F	T	$0.29 \times 0.9 \times 0.7 = 0.182700$	$0.71 \times 0.05 \times 0.01 = 0.000355$	0.183055
F	F	$0.001 \times 0.9 \times 0.7 = 0.000630$	$0.999 \times 0.05 \times 0.01 = 0.000500$	0.001130

For $B = T$:

$$\begin{aligned}
 P(B=T, J, M) &= \alpha \times 0.001 \times [0.002 \times 0.598525 + 0.998 \times 0.592230] \\
 &= \alpha \times 0.001 \times [0.001197 + 0.591046] \\
 &= \alpha \times 0.001 \times 0.592243 = 0.000592 \alpha
 \end{aligned}$$

For $B = F$:

$$\begin{aligned}
 P(B=F, J, M) &= \alpha \times 0.999 \times [0.002 \times 0.183055 + 0.998 \times 0.001130] \\
 &= \alpha \times 0.999 \times [0.000366 + 0.001128] \\
 &= \alpha \times 0.999 \times 0.001494 = 0.001493 \alpha
 \end{aligned}$$

Normalization:

$$\alpha = \frac{1}{0.000592 + 0.001493} = \frac{1}{0.002085} \approx 479.6$$

$$P(B=T \mid J, M) = 479.6 \times 0.000592 \approx 0.284$$

$$P(B=F \mid J, M) = 479.6 \times 0.001493 \approx 0.716$$

$$\boxed{P(\text{Burglary} \mid J=T, M=T) = \langle 0.284, 0.716 \rangle}$$

Given that both John and Mary called, there is approximately a **28.4% chance a burglary occurred**, and a 71.6% chance it did not. This matches the result from the textbook.

(b) Chain network: enumeration vs. variable elimination

Consider a chain $X_1 \rightarrow X_2 \rightarrow \dots \rightarrow X_n$ where we compute $P(X_1 \mid X_n = T)$.

Enumeration. We must sum over all 2^{n-2} joint assignments to $X_2, \dots, X_{n-1}$. Each such assignment requires n multiplications (one per factor in the chain), so the total cost is

$$O(n \cdot 2^{n-2}) = O(n \cdot 2^n),$$

which is **exponential** in n . More concretely, for n variables there are 2^{n-2} hidden-variable assignments, and each requires $(n-1)$ multiplications plus one for the evidence factor, giving approximately $n \cdot 2^{n-2}$ multiplications and $2^{n-2} - 1$ additions in the summation.

Variable elimination. We eliminate variables right-to-left. Starting from the rightmost hidden variable X_{n-1} :

$$P(X_1 \mid X_n=T) = \alpha P(X_1) \cdots \sum_{x_{n-2}} P(x_{n-2} \mid x_{n-3}) \underbrace{\sum_{x_{n-1}} P(x_{n-1} \mid x_{n-2}) P(X_n=T \mid x_{n-1})}_{f_{X_{n-1}}(x_{n-2})}$$

Summing out X_{n-1} (a binary variable) costs:

- 2 multiplications (one per value of x_{n-1}),
- 1 addition, and
- produces a factor $f_{X_{n-1}}$ over the single variable x_{n-2} .

Each of the $n-2$ elimination steps has the same cost, so the total number of arithmetic operations is:

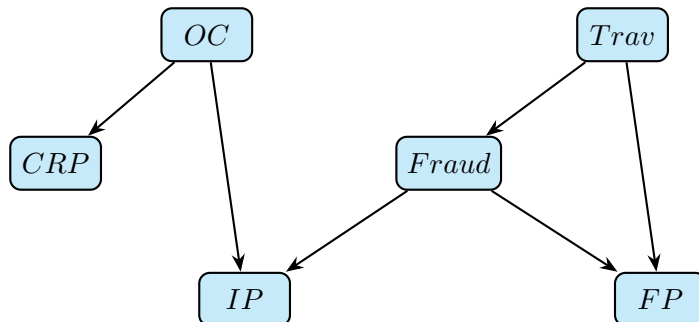
$$(n-2) \times (2 \text{ multiplications} + 1 \text{ addition}) + 1 \text{ normalisation division} = O(n),$$

which is **linear** in n — exponentially faster than enumeration.

3 Problem 3 (20 points): Fraud Detection

(a) Bayesian Network

$Trav$ and OC are root nodes. $Fraud$ depends on $Trav$. FP depends on $Trav$ and $Fraud$. IP depends on $Fraud$ and OC . CRP depends on OC .



$$P(Trav=T) = 0.05, \quad P(OC=T) = 0.75$$

$Trav$	$P(Fraud=T)$	$P(Fraud=F)$
T	0.010	0.990
F	0.004	0.996

$Trav$	$Fraud$	$P(FP=T)$	$P(FP=F)$	$Fraud$	OC	$P(IP=T)$	$P(IP=F)$
T	T	0.90	0.10	T	T	0.020	0.980
T	F	0.90	0.10	T	F	0.011	0.989
F	T	0.10	0.90	F	T	0.010	0.990
F	F	0.01	0.99	F	F	0.001	0.999

OC	$P(CRP=T)$	$P(CRP=F)$
T	0.100	0.900
F	0.001	0.999

Note: The FP CPT has $P(FP=T \mid Trav=T) = 0.90$ for both values of $Fraud$, because the problem states “when the card holder is travelling, 90% of transactions are foreign purchases regardless of the legitimacy of the transactions.”

(b) Probability Queries

Query 1: $P(Fraud)$

$$\begin{aligned}
 P(Fraud=T) &= P(Fraud=T \mid Trav=T) \cdot P(Trav=T) + P(Fraud=T \mid Trav=F) \cdot P(Trav=F) \\
 &= 0.01 \times 0.05 + 0.004 \times 0.95 = 0.0005 + 0.0038 = \boxed{0.0043}
 \end{aligned}$$

Before any transaction evidence is observed, the **baseline probability of fraud is 0.43%**.

Query 2: $P(\textit{Fraud} \mid \textit{FP}=T, \textit{IP}=F, \textit{CRP}=T)$ With $\alpha = \frac{1}{P(\textit{FP}=T, \textit{IP}=F, \textit{CRP}=T)}$, marginalizing over $\textit{Trav}$ and $\textit{OC}$:

$$P(\textit{Fraud} \mid \textit{FP}, \neg \textit{IP}, \textit{CRP}) = \alpha \sum_{\textit{Trav}} P(\textit{Trav}) P(\textit{Fraud} \mid \textit{Trav}) P(\textit{FP}=T \mid \textit{Trav}, \textit{Fraud}) \\ \times \sum_{\textit{OC}} P(\textit{OC}) P(\textit{CRP}=T \mid \textit{OC}) P(\textit{IP}=F \mid \textit{Fraud}, \textit{OC})$$

First we compute the inner OC sum for each value of Fraud:

$$\text{OC sum (Fraud=T)} = 0.75 \times 0.1 \times 0.980 + 0.25 \times 0.001 \times 0.989 = 0.07350 + 0.000247 = 0.073747$$

$$\text{OC sum (Fraud=F)} = 0.75 \times 0.1 \times 0.990 + 0.25 \times 0.001 \times 0.999 = 0.07425 + 0.000250 = 0.074500$$

For $\textit{Fraud} = T$:

$$= \alpha \times \left[\underbrace{0.05 \times 0.01 \times 0.90}_{\textit{Trav}=T} + \underbrace{0.95 \times 0.004 \times 0.10}_{\textit{Trav}=F} \right] \times 0.073747 \\ = \alpha \times (0.000450 + 0.000380) \times 0.073747 \\ = \alpha \times 0.000830 \times 0.073747 \\ = 0.0000612 \alpha$$

For $\textit{Fraud} = F$:

$$= \alpha \times \left[\underbrace{0.05 \times 0.99 \times 0.90}_{\textit{Trav}=T} + \underbrace{0.95 \times 0.996 \times 0.01}_{\textit{Trav}=F} \right] \times 0.074500 \\ = \alpha \times (0.044550 + 0.009462) \times 0.074500 \\ = \alpha \times 0.054012 \times 0.074500 \\ = 0.004024 \alpha$$

Normalization:

$$\alpha = \frac{1}{0.0000612 + 0.004024} = \frac{1}{0.004085} \approx 244.8$$

$$P(\textit{Fraud}=T \mid \textit{FP}=T, \textit{IP}=F, \textit{CRP}=T) = 244.8 \times 0.0000612 \approx 0.015$$

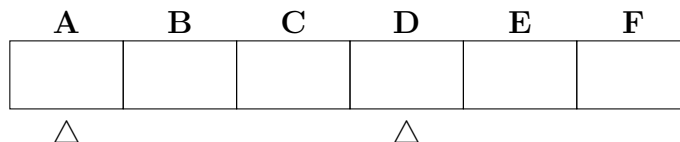
$$P(\textit{Fraud}=F \mid \textit{FP}=T, \textit{IP}=F, \textit{CRP}=T) = 244.8 \times 0.004024 \approx 0.985$$

$$\boxed{P(\textit{Fraud} \mid \textit{FP}=T, \textit{IP}=F, \textit{CRP}=T) = \langle 0.015, 0.985 \rangle}$$

The probability that the transaction is fraudulent **increases from 0.43% to approximately 1.5%** given the evidence of a foreign purchase, no internet purchase, and a recent computer-related accessory purchase. While the foreign purchase raises suspicion, the overall fraud probability remains low because non-fraudulent travellers are also likely to make foreign purchases.

4 Problem 4 (40 points): HMM Search and Rescue

The rover starts at position A on day 1 ($X_1 = A$). Positions are $X_t \in \{A, B, C, D, E, F\}$; the rover moves east with probability 0.8 and stays with probability 0.2. Volcanic vents at A and D give $P(E_t=\text{hot} \mid X_t \in \{A, D\}) = 1$ and $P(E_t=\text{cold} \mid X_t \notin \{A, D\}) = 1$ with no noise.



Transition model $P(X_{t+1} \mid X_t)$: Each state transitions east with $p = 0.8$ or stays with $p = 0.2$; F is absorbing.

	A	B	C	D	E	F
A	0.2	0.8	0	0	0	0
B	0	0.2	0.8	0	0	0
C	0	0	0.2	0.8	0	0
D	0	0	0	0.2	0.8	0
E	0	0	0	0	0.2	0.8
F	0	0	0	0	0	1

Observation model $P(E_t \mid X_t)$:

X_t	$P(\text{hot})$	$P(\text{cold})$
A	1	0
B	0	1
C	0	1
D	1	0
E	0	1
F	0	1

The forward filtering recurrence is:

$$P(X_t \mid e_{1:t}) \propto P(E_t \mid X_t) \sum_{X_{t-1}} P(X_t \mid X_{t-1}) P(X_{t-1} \mid e_{1:t-1})$$

Day 1. $X_1 = A$ with certainty; $E_1 = \text{hot}$ is consistent with A being a vent:

$$P(X_1 \mid E_1=\text{hot}) = \langle 1, 0, 0, 0, 0, 0 \rangle$$

Day 2 (intermediate computation). Propagating forward from $X_1 = A$:

$$\sum_{X_1} P(X_2 \mid X_1) P(X_1 \mid e_1) = \langle 0.2, 0.8, 0, 0, 0, 0 \rangle$$

Multiplying element-wise by $P(E_2=\text{cold} \mid X_2) = \langle 0, 1, 1, 0, 1, 1 \rangle$:

$$\propto \langle 0, 0.8, 0, 0, 0, 0 \rangle \xrightarrow{\times 1.25} \langle 0, 1, 0, 0, 0, 0 \rangle$$

$$P(X_2 \mid e_{1:2}) = \langle 0, 1.0, 0, 0, 0, 0 \rangle$$

1. Filtering: $P(X_3 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3)$

Propagating forward from $X_2 = B$ with certainty:

$$\sum_{X_2} P(X_3 \mid X_2) P(X_2 \mid e_{1:2}) = \langle 0, 0.2, 0.8, 0, 0, 0 \rangle$$

Multiplying element-wise by $P(E_3=\text{cold} \mid X_3) = \langle 0, 1, 1, 0, 1, 1 \rangle$:

$$\propto \langle 0, 0.2, 0.8, 0, 0, 0 \rangle$$

Values sum to 1.0; no further normalization needed.

X_3	$P(X_3 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3)$
A	0.0
B	0.2
C	0.8
D	0.0
E	0.0
F	0.0

$$P(X_3 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3) = \langle 0, 0.2, 0.8, 0, 0, 0 \rangle$$

After three days of observations (hot, cold, cold), the rover is most likely at position **C** with probability **0.8**, or at B with probability 0.2.

2. Smoothing: $P(X_2 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3)$

Using the forward-backward algorithm: $P(X_2 \mid e_{1:3}) \propto P(X_2 \mid e_{1:2}) \times P(e_3 \mid X_2)$

First compute the backward message $P(E_3=\text{cold} \mid X_2)$ by summing over X_3 :

$$P(E_3=\text{cold} \mid X_2) = \sum_{X_3} P(X_3 \mid X_2) P(E_3=\text{cold} \mid X_3)$$

X_2	Computation
A	$0.2 \times 0 + 0.8 \times 1 = 0.8$
B	$0.2 \times 1 + 0.8 \times 1 = 1.0$
C	$0.2 \times 1 + 0.8 \times 0 = 0.2$
D	$0.2 \times 0 + 0.8 \times 1 = 0.8$
E	$0.2 \times 1 + 0.8 \times 1 = 1.0$
F	$0 \times 1 + 1 \times 1 = 1.0$

So the backward message is $\langle 0.8, 1.0, 0.2, 0.8, 1.0, 1.0 \rangle$.

Since $P(X_2 | e_{1:2}) = \langle 0, 1, 0, 0, 0, 0 \rangle$ (nonzero only at B):

$$P(X_2 | e_{1:3}) \propto \langle 0.8, 1.0, 0.2, 0.8, 1.0, 1.0 \rangle \odot \langle 0, 1, 0, 0, 0, 0 \rangle = \langle 0, 1.0, 0, 0, 0, 0 \rangle$$

X_2	$P(X_2 \text{hot}_1, \text{cold}_2, \text{cold}_3)$
A	0.0
B	1.0
C	0.0
D	0.0
E	0.0
F	0.0

$$P(X_2 | \text{hot}_1, \text{cold}_2, \text{cold}_3) = \langle 0, 1.0, 0, 0, 0, 0 \rangle$$

Using all three days of evidence, we can confirm with **certainty that the rover was at position B on day 2**.

3. Prediction: $P(\text{hot}_4 | \text{hot}_1, \text{cold}_2, \text{cold}_3)$

We first need $P(X_4 | e_{1:3})$ (computed in Part 4 below), then:

$$P(\text{hot}_4 | e_{1:3}) = \sum_{X_4} P(E_4=\text{hot} | X_4) P(X_4 | e_{1:3})$$

Only $X_4 = D$ contributes since $P(\text{hot} | X_4) = 1$ only when $X_4 \in \{A, D\}$, and from Part 4, $P(X_4=D | e_{1:3}) = 0.64$:

$$= 0 \times 0.04 + 0 \times 0.32 + 1 \times 0.64 + 0 \times 0 + 0 \times 0 = \boxed{0.64}$$

There is a **64% probability of observing a hot reading on day 4**, corresponding to the rover having reached vent D.

4. Prediction: $P(X_4 | \text{hot}_1, \text{cold}_2, \text{cold}_3)$

Propagating $P(X_3 | e_{1:3})$ forward one step with no new evidence:

$$P(X_4 | e_{1:3}) = \sum_{X_3} P(X_4 | X_3) P(X_3 | e_{1:3})$$

$$P(X_4=A) = 0 \quad (\text{unreachable})$$

$$P(X_4=B) = P(B | B) \times 0.2 + P(B | C) \times 0.8 = 0.2 \times 0.2 + 0 \times 0.8 = 0.04$$

$$P(X_4=C) = P(C | B) \times 0.2 + P(C | C) \times 0.8 = 0.8 \times 0.2 + 0.2 \times 0.8 = 0.32$$

$$P(X_4=D) = P(D | B) \times 0.2 + P(D | C) \times 0.8 = 0 \times 0.2 + 0.8 \times 0.8 = 0.64$$

$$P(X_4=E) = 0 \quad (\text{unreachable in one step from B or C})$$

$$P(X_4=F) = 0$$

X_4	$P(X_4 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3)$
A	0.00
B	0.04
C	0.32
D	0.64
E	0.00
F	0.00

$$P(X_4 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3) = \langle 0, 0.04, 0.32, 0.64, 0, 0 \rangle$$

On day 4, the rover is most likely at **position D with probability 0.64**, having traveled east from C (which itself was the most likely position on day 3).

Bonus (10 points): $P(\text{hot}_4, \text{hot}_5, \text{cold}_6 \mid \text{hot}_1, \text{cold}_2, \text{cold}_3)$

We use the likelihood of future evidence given current state, working backwards:

$$P(e_4, e_5, e_6 \mid e_{1:3}) = \sum_{X_3} P(X_3 \mid e_{1:3}) P(e_{4:6} \mid X_3)$$

where $P(e_{4:6} \mid X_3) = \sum_{X_4} P(X_4 \mid X_3) P(e_4 \mid X_4) P(e_{5:6} \mid X_4)$.

Step 1 — Compute $P(E_6=\text{cold} \mid X_5)$:

$$P(E_6=\text{cold} \mid X_5) = \sum_{X_6} P(X_6 \mid X_5) P(\text{cold} \mid X_6)$$

X_5	Computation
A	$0.2 \times P(\text{cold} \mid A) + 0.8 \times P(\text{cold} \mid B) = 0.2 \times 0 + 0.8 \times 1 = 0.8$
B	$0.2 \times 1 + 0.8 \times 1 = 1.0$
C	$0.2 \times 1 + 0.8 \times 0 = 0.2$
D	$0.2 \times 0 + 0.8 \times 1 = 0.8$
E	$0.2 \times 1 + 0.8 \times 1 = 1.0$
F	$1 \times 1 = 1.0$

$$P(E_6=\text{cold} \mid X_5) = \langle 0.8, 1.0, 0.2, 0.8, 1.0, 1.0 \rangle$$

Step 2 — Compute $P(E_5=\text{hot}, E_6=\text{cold} \mid X_4)$:

$$P(e_{5:6} \mid X_4) = \sum_{X_5} P(X_5 \mid X_4) P(\text{hot} \mid X_5) P(E_6=\text{cold} \mid X_5)$$

X_4	Computation
A	$0.2 \times P(\text{hot} \mid A) \times 0.8 + 0.8 \times P(\text{hot} \mid B) \times 1.0 = 0.2 \times 1 \times 0.8 + 0.8 \times 0 \times 1.0 = 0.16$
B	$0.2 \times 0 \times 1.0 + 0.8 \times 0 \times 0.2 = 0$
C	$0.2 \times 0 \times 0.2 + 0.8 \times 1 \times 0.8 = 0.64$
D	$0.2 \times 1 \times 0.8 + 0.8 \times 0 \times 1.0 = 0.16$
E	$0.2 \times 0 + 0.8 \times 0 = 0$
F	0

$$P(e_{5:6} \mid X_4) = \langle 0.16, 0, 0.64, 0.16, 0, 0 \rangle$$

Step 3 — Compute $P(E_4=\text{hot}, E_5=\text{hot}, E_6=\text{cold} \mid X_3)$:

$$P(e_{4:6} \mid X_3) = \sum_{X_4} P(X_4 \mid X_3) P(\text{hot} \mid X_4) P(e_{5:6} \mid X_4)$$

X_3	Computation
B	$0.2 \times P(\text{hot} \mid B) \times 0 + 0.8 \times P(\text{hot} \mid C) \times 0.64 = 0$
C	$0.2 \times P(\text{hot} \mid C) \times 0.64 + 0.8 \times P(\text{hot} \mid D) \times 0.16 = 0.2 \times 0 \times 0.64 + 0.8 \times 1 \times 0.16 = 0.128$

$$P(e_{4:6} \mid X_3) = \langle 0, 0, 0.128, 0, 0, 0 \rangle \quad (\text{only nonzero at } X_3=C)$$

Final combination:

$$\begin{aligned}
P(\text{hot}_4, \text{hot}_5, \text{cold}_6 \mid e_{1:3}) &= \sum_{X_3} P(X_3 \mid e_{1:3}) P(e_{4:6} \mid X_3) \\
&= 0 \times 0 + 0.2 \times 0 + 0.8 \times 0.128 + 0 \times 0 + 0 \times 0 + 0 \times 0 \\
&= 0.8 \times 0.128 = \boxed{0.1024}
\end{aligned}$$

The probability of observing hot, hot, cold on days 4, 5, and 6 respectively is **10.24%**. This requires the rover to reach vent D by day 4 (probability 0.64), stay at D on day 5 (probability 0.2), and then move to E on day 6 (probability 0.8).

5 Problem 5 (10 points): Markov Decision Process

States $\{A, B\}$, actions $\{1, 2\}$, discount $\gamma = 1$:

s	a	s'	$T(s, a, s')$		s	a	$R(s, a)$
A	1	A	1.0		A	1	0
A	1	B	0.0		A	2	-1
A	2	A	0.5		B	1	5
A	2	B	0.5		B	2	0
B	1	A	0.0				
B	1	B	1.0				
B	2	A	0.0				
B	2	B	1.0				

1. Bellman Equation

$$V^\pi(s) = R(s, \pi(s)) + \gamma \sum_{s'} T(s, \pi(s), s') V^\pi(s')$$

With $\gamma = 1$ this simplifies to:

$$V^\pi(s) = R(s, \pi(s)) + \sum_{s'} T(s, \pi(s), s') V^\pi(s')$$

2. Two Iterations of Policy Evaluation ($\pi(A) = 1, \pi(B) = 1$)

Initialize $V_0^\pi(A) = V_0^\pi(B) = 0$.

Iteration 1:

$$V_1^\pi(A) = R(A, 1) + T(A, 1, A) V_0^\pi(A) + T(A, 1, B) V_0^\pi(B) = 0 + 1 \times 0 + 0 \times 0 = 0$$

$$V_1^\pi(B) = R(B, 1) + T(B, 1, A) V_0^\pi(A) + T(B, 1, B) V_0^\pi(B) = 5 + 0 \times 0 + 1 \times 0 = 5$$

Iteration 2:

$$V_2^\pi(A) = R(A, 1) + T(A, 1, A) V_1^\pi(A) + T(A, 1, B) V_1^\pi(B) = 0 + 1 \times 0 + 0 \times 5 = \boxed{0}$$

$$V_2^\pi(B) = R(B, 1) + T(B, 1, A) V_1^\pi(A) + T(B, 1, B) V_1^\pi(B) = 5 + 0 \times 0 + 1 \times 5 = \boxed{10}$$

After two iterations, $V_2^\pi(A) = 0$ and $V_2^\pi(B) = 10$.

3. Policy Improvement

$$\pi_{\text{new}}(s) = \arg \max_a \left[R(s, a) + \sum_{s'} T(s, a, s') V_2^\pi(s') \right]$$

State A:

$$a = 1: \quad 0 + 1 \times 0 + 0 \times 10 = 0$$

$$a = 2: \quad -1 + 0.5 \times 0 + 0.5 \times 10 = 4$$

Since $4 > 0$, set $\boxed{\pi_{\text{new}}(A) = 2}$.

State B :

$$a = 1 : \quad 5 + 0 \times 0 + 1 \times 10 = 15$$

$$a = 2 : \quad 0 + 0 \times 0 + 1 \times 10 = 10$$

Since $15 > 10$, set $\boxed{\pi_{\text{new}}(B) = 1}$.

$$\boxed{\pi_{\text{new}}(A) = 2, \quad \pi_{\text{new}}(B) = 1}$$

The improved policy **switches state A from action 1 to action 2**: staying at A forever (action 1) yields value 0, whereas action 2 gives a 50% chance of reaching B each step, which has value 10, making the expected return $-1 + 0.5 \times 10 = 4 > 0$. State B retains action 1, as it yields $5 + 10 = 15$ versus action 2's $0 + 10 = 10$.